NGUYEN XUAN TRUNG

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AI Engineer

EDUCATION

FPT University 2023 - Present
Bachelor of Science in Artificial Intelligence Expected: 06/2027
GPA: 3.75/4.00 | **English:** B2

WORK EXPERIENCE

ECE Technology Co., Ltd. | AI Engineer Intern 01/2026 - 05/2026
Project: AI for E-Commerce Platform (MVP v1)

- Built an **RFM-based product segmentation** module using Mini-Batch K-Means with 7-day sales velocity and 30-day labels.
- Developed a **Gemini and SerpAPI** workflow for market research and structured business recommendations.
- Built **FastAPI** and PostgreSQL endpoints; Dockerized and deployed the service to **AWS EC2**.

PROJECTS

TraceVision - Evidence-Guided Video Investigation | Sole developer 08/2026 - 09/2026
Python, FastAPI, LangGraph, YOLO, faster-whisper, OpenCLIP, FAISS, SQLite

- Built a **multimodal video search** pipeline (detection, transcription, visual embeddings) answering Vietnamese and English queries with timestamped evidence.
- Routed queries between retrieval and VLM calls with **LangGraph**; achieved a **52.3× indexing speedup** via dependency-aware caching.

Subject Knowledge Hub | Sole developer 08/2026 - 09/2026
React, TypeScript, FastAPI, PostgreSQL/pgvector, Celery, Redis, OpenAI API, Docker Compose

- Built a full-stack **PDF RAG** app with authentication, document versioning, page-level citations, and resumable ingestion (Celery, Redis).
- Reduced input tokens by **25.3%** with page-aligned chunking while keeping **100% page Recall@6** on 56 answerable questions.

Secure Login System | Project leader 08/2025 - 11/2025
Python, PyTorch, OpenCV, MTCNN, ONNX

- Developed a **CDCN-based face anti-spoofing** module in PyTorch (spatial attention, CBAM) with training and evaluation pipelines.
- Built an OpenCV and MTCNN webcam pipeline; exported the CBAM model to **ONNX**.

PUBLICATIONS & HONORS

- *Optimizing YOLOv11n for Real-Time Object Detection: Leveraging Quantization and Model Optimization.* **EIDT 2025**, Springer LNEE, vol. 1531, 2026. DOI: 10.1007/978-981-95-6111-7_13
- *Weather Forecasting System "Four Seasons in One Day" and Shelter Suggestion for Sydney, Melbourne and Canberra.* **AJCAI 2025**, Springer LNAI, vol. 16370, 2026. DOI: 10.1007/978-981-95-4969-6_33
- **Top 100 Excellent Students**, FPT University (Summer & Fall Semesters), 2025.

SKILLS

Programming & ML: Python, SQL, PyTorch, TensorFlow, scikit-learn, OpenCV
AI & LLM: Computer Vision, NLP, Multimodal Learning, RAG, LangChain, LangGraph
Backend & Tools: FastAPI, REST APIs, PostgreSQL, Docker, AWS EC2, Git